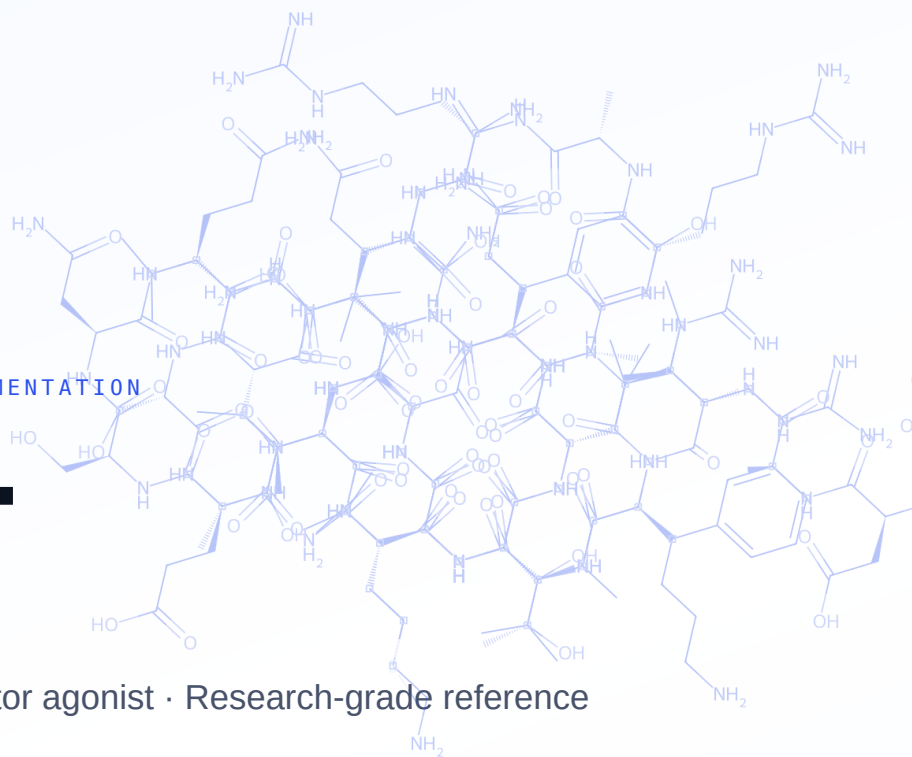


● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

GLP-RT

Triple GIP / GLP-1 / glucagon receptor agonist · Research-grade reference material



COMPOUND

Retatrutide

CLASS

Triple incretin agonist

STATUS

Investigational

MW

≈4731 g/mol

Sold for laboratory research purposes only. Not for human or veterinary use, consumption, or therapeutic application. Must be 21+ to purchase. | Independently verified · HPLC + mass spectrometry · Certificate of Analysis per lot.

Identity & Composition

Retatrutide is a synthetic peptide built on a GIP-analogue backbone with a fatty-acid modification supporting once-weekly pharmacokinetics. It is a first-in-class **triple** hormone receptor agonist — a single molecule activating GIP, GLP-1 and glucagon receptors. The glucagon component is the mechanistic addition relative to dual agonists.

Product name	GLP-RT (retatrutide)
Class	Triple GIP / GLP-1 / glucagon receptor agonist
Backbone	GIP-analogue synthetic peptide
Length	39 amino acids
Modification	Fatty-acid conjugate (half-life extension)
Molecular weight	≈4731 g/mol
Development status	Investigational — Phase 3 (TRIUMPH programme)
Appearance	White lyophilized powder
Identified use	In-vitro research & development only

REGULATORY STATUS

Retatrutide is investigational and is not approved by the FDA, EMA, or any other regulatory authority for any indication.

Its complete safety profile has not been established, and Phase 3 evaluation remains ongoing. Supplied solely as a reference material for in-vitro laboratory research; not for human or veterinary administration under any circumstances.

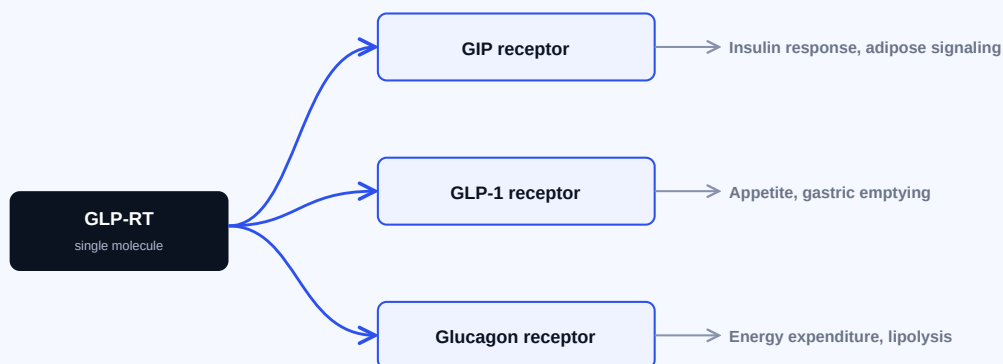
Why the third receptor matters scientifically. GIP and GLP-1 agonism act largely on appetite, gastric emptying and insulin response. Glucagon receptor agonism adds a distinct signal associated with energy expenditure and lipolysis — which is the mechanistic rationale for studying triple agonism, and also why its safety profile cannot be assumed from dual-agonist experience.

Mechanism

Retatrutide extends incretin co-agonism to a third target. Where dual agonists engage the appetite and insulin axes, adding glucagon receptor agonism introduces a direct energy-expenditure and lipolytic signal — a mechanistically different contribution rather than simply more of the same.

Pharmacology

- **GIP receptor agonism.** Incretin signaling; insulin response and adipose-tissue effects.
- **GLP-1 receptor agonism.** Appetite reduction and slowed gastric emptying, as in the established GLP-1 class.
- **Glucagon receptor agonism.** Associated with increased energy expenditure and fat metabolism — the differentiating component.
- **Once-weekly pharmacokinetics.** Fatty-acid modification supports extended half-life in the investigational clinical formulation.
- **Open scientific question.** Whether the added glucagon signal alters the long-term safety profile relative to dual agonism is precisely what the ongoing Phase 3 programme is designed to determine.



Triple hormone receptor co-agonism. Schematic. Research-grade material supplied here is for in-vitro laboratory use only.

What the Literature Supports — and What It Doesn't

Retatrutide has produced striking Phase 3 topline results. It is nonetheless **investigational**: no regulatory authority has approved it, most Phase 3 readouts are pending, and much of the currently public data is company topline rather than full peer-reviewed publication.

Evidence at a glance

PHASE 3

Ongoing (TRIUMPH)

TOPLINE RESULTS

Positive

REGULATORY APPROVAL

None — anywhere

LONG-TERM SAFETY

Not established

- **TRIUMPH-1** (NCT05929066; N=2,339, 80 weeks, obesity/overweight without diabetes): all doses (4 / 9 / 12 mg) met primary and key secondary endpoints, with placebo-adjusted mean weight reduction reported across doses versus 2.2% placebo. Topline announced May 2026.
- **TRIUMPH-4** (obesity with knee osteoarthritis, 68 weeks): met co-primary endpoints for weight and pain/physical function. Topline announced December 2025.
- **Phase 2** data published 2023 supported progression to Phase 3.
- **Seven additional Phase 3 readouts were expected during 2026**, spanning type 2 diabetes, OSA, chronic low back pain, cardiovascular/renal outcomes and MASLD. The full picture is not yet complete.
- **Much public data remains topline / conference-presented** rather than fully peer-reviewed, and independent scrutiny of complete datasets is limited.

Appropriate characterization: a first-in-class triple agonist with strong but still-accumulating late-stage clinical evidence, no regulatory approval in any jurisdiction, and an incompletely characterized long-term safety profile. Information here is current as of the revision date; the TRIUMPH programme is actively reading out and this section should be re-verified before reuse.

Regulatory & status notes

US FDA	Not approved. Investigational; Phase 3 ongoing.
Other jurisdictions	Not approved by EMA or any other regulatory authority.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Jastreboff AM, et al. Triple-hormone-receptor agonist retatrutide for obesity — a Phase 2 trial. *N Engl J Med*. 2023.
2. Eli Lilly and Company. TRIUMPH-1 Phase 3 topline results (NCT05929066). May 2026.
3. Eli Lilly and Company. TRIUMPH-4 Phase 3 topline results, obesity with knee osteoarthritis. December 2025.
4. TRANSCEND-T2D-1 and TRIUMPH-1 results, ADA Scientific Sessions 2026.

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	GLP-RT (retatrutide)
Class	Triple GIP / GLP-1 / glucagon agonist, 39 aa
MW	≈4731 g/mol
Status	Investigational — not approved
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use. Research-grade material is not an approved drug product.
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data for the research-grade substance; occupational toxicological data are limited. **Signal word:** None. **Pictograms:** None required.

Precautionary: P261, P264, P280, P501. **Handle as a potent pharmacologically active peptide** — active at microgram-to-milligram quantities in vivo; minimize all exposure.

3 Composition

Single-substance product, >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** fresh air; seek medical advice if symptoms develop. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Inform medical personnel that the substance is an incretin/glucagon receptor agonist; monitor for glycaemic and gastrointestinal effects.

5 Fire-Fighting

Media: CO₂, dry chemical, foam, water spray. Combustion may produce CO, CO₂, nitrogen oxides and sulfur oxides. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust formation; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13. Decontaminate the affected surface.

7 Handling & Storage

Handle as a potent pharmacologically active peptide; containment is important. Weigh and reconstitute in a fume hood or ventilated balance enclosure. Store lyophilized solid tightly closed, protected from light/moisture/heat; -20 °C long-term, 2–8 °C short-term. Aliquot reconstituted solutions; store frozen; avoid repeated freeze-thaw. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Because the substance is pharmacologically potent and its safety profile is not fully characterized, engineering containment when handling dry powder is a priority. Fume hood or ventilated balance enclosure; nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator where controls are insufficient.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Soluble in water
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable as a lyophilized solid under recommended storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents. Decomposition: CO, CO₂, NO_x, SO_x. No hazardous polymerization.

11 Toxicological Information

The full toxicological and safety profile of this investigational substance has not been established.

Clinical safety data from ongoing trials pertain to the investigational drug product under medical supervision and do not characterize occupational exposure to research-grade material. No occupational LD₅₀/LC₅₀ identified. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

Retatrutide is an investigational compound and is not approved by the FDA, EMA or any other regulatory authority for any indication. Material supplied here is research grade, is not an approved or investigational drug product, is not manufactured to pharmaceutical standards, and must not be represented, dispensed or used as a medicine. US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use. Note that the development status of this investigational compound may change; re-verify before reuse. Sources: PubChem; sponsor trial reports; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8. Provided without warranty. For scientific R&D use only.

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